
*An Integrated Computational Pipeline for Virtual Screening
of Natural Compounds Against Alzheimer's Disease Using
Molecular Docking and Python-Based Data Analytics*

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Computational Drug Discovery | Molecular Docking | Python |
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Abstract

Background: Alzheimer's disease (AD) is a progressive neurodegenerative disorder characterized by cognitive decline and cholinergic dysfunction. Acetylcholinesterase (AChE) remains a well-established therapeutic target for symptomatic management of AD. Computational drug discovery approaches enable the rapid screening and prioritization of potential bioactive compounds prior to experimental validation.

Methods: An integrated computational workflow was developed to evaluate 41 natural compounds against human acetylcholinesterase (PDB ID: 4EY7). Ligands were selected through literature review and retrieved from the PubChem database, while the target protein was obtained from the Protein Data Bank. Molecular docking was performed using AutoDock Vina implemented in PyRx. The docking results were further analyzed using Python-based data analytics, including molecular descriptor evaluation, descriptive statistical analysis, and data visualization. Protein–ligand interactions were examined using Discovery Studio Visualizer, and drug-likeness and pharmacokinetic properties were assessed using Lipinski's Rule of Five and the SwissADME web server.

Results: Molecular docking identified epigallocatechin gallate (EGCG) as the highest-ranked compound, exhibiting a predicted binding affinity of -11.97 kcal/mol, slightly higher than the reference inhibitor donepezil (-11.77 kcal/mol). Demethoxycurcumin, curcumin, and bisdemethoxycurcumin also demonstrated favorable predicted binding affinities. Drug-likeness and ADMET analyses indicated that although EGCG exhibited the strongest predicted binding, several other compounds possessed more favorable pharmacokinetic characteristics for potential central nervous system drug development.

Conclusion: The integrated computational pipeline successfully prioritized several natural compounds with promising predicted interactions against human acetylcholinesterase. These findings provide a rational basis for future molecular dynamics simulations and experimental validation while demonstrating the utility of combining molecular docking, Python-based data analysis, and pharmacokinetic prediction for early-stage lead identification.

Keywords: Alzheimer's disease; Acetylcholinesterase; Molecular docking; Virtual screening; Natural compounds; Drug-likeness; ADMET prediction; Computational drug discovery; Python data analysis.

1.Introduction

1.1 Alzheimer's disease: Background and significance

Alzheimer's disease (AD) is the most common cause of dementia and a progressive neurodegenerative disorder characterized by gradual impairment of memory, cognition, and behavior. The disease is associated with the accumulation of extracellular amyloid- β plaques, intracellular neurofibrillary tangles composed of hyperphosphorylated tau protein, synaptic dysfunction, and progressive neuronal loss. As the global population continues to age, the prevalence of Alzheimer's disease is expected to increase substantially, posing significant medical, social, and economic challenges worldwide. Despite decades of research, currently available therapies provide only symptomatic relief and do not effectively halt or reverse disease progression. Consequently, the discovery of safer and more effective therapeutic agents remains a major priority in Alzheimer's disease research.

1.2 Acetylcholinesterase as a Therapeutic Target

One of the major pathological features of Alzheimer's disease is the progressive degeneration of cholinergic neurons, resulting in reduced levels of the neurotransmitter acetylcholine in the brain. According to the cholinergic hypothesis, this decline in cholinergic neurotransmission contributes significantly to cognitive impairment and memory loss observed in patients with Alzheimer's disease. Acetylcholinesterase (AChE), the enzyme responsible for the rapid hydrolysis of acetylcholine within the synaptic cleft, has therefore become a well-established therapeutic target for the symptomatic management of the disease. Inhibition of AChE increases the availability of acetylcholine at neuronal synapses, thereby enhancing cholinergic neurotransmission and temporarily improving cognitive function. Currently approved AChE inhibitors, including donepezil, rivastigmine, and galantamine, have demonstrated clinical benefits in improving cognitive symptoms. However, their therapeutic efficacy is limited by factors such as adverse effects, variable patient response, and their inability to prevent or reverse the underlying neurodegenerative process. These limitations highlight the need to identify novel acetylcholinesterase inhibitors with improved efficacy, safety, and pharmacokinetic properties.

1.3 Natural Compounds as Potential Sources of Acetylcholinesterase Inhibitors

Natural products have long served as an invaluable source of therapeutic agents due to their remarkable structural diversity and broad spectrum of biological activities. A significant proportion of clinically approved drugs are either directly derived from natural products or developed from naturally occurring chemical scaffolds, highlighting their importance in modern drug discovery. Numerous plant-derived phytochemicals, including flavonoids, polyphenols, alkaloids, terpenoids, and curcuminoids, have been reported to possess antioxidant, anti-inflammatory, neuroprotective, and acetylcholinesterase inhibitory properties. These biological activities are particularly relevant to Alzheimer's disease, where oxidative stress, chronic neuroinflammation, and cholinergic dysfunction contribute to disease progression. Furthermore, many natural compounds exhibit multi-target pharmacological effects, allowing them to modulate multiple pathological pathways simultaneously rather than acting on a single molecular target. Consequently, natural products represent a promising reservoir for the discovery of novel acetylcholinesterase inhibitors with the potential to complement or improve existing therapeutic strategies for Alzheimer's disease.

1.4 Role of Computational Drug Discovery and Virtual Screening

Advances in computational drug discovery have significantly accelerated the identification and optimization of potential therapeutic candidates by enabling the rapid evaluation of large numbers of compounds prior to experimental investigation. Among these approaches, molecular docking has become one of the most widely employed techniques for predicting the binding orientation and binding affinity of small molecules within the active site of target proteins. Although molecular docking does not directly measure biological activity, it provides valuable insights into protein–ligand interactions and facilitates the prioritization of compounds for further study. Complementary computational analyses, including molecular descriptor evaluation, statistical data analysis, drug-likeness assessment, and absorption, distribution, metabolism, excretion, and toxicity (ADMET) prediction, further improve the lead selection process by assessing the physicochemical and pharmacokinetic properties of candidate molecules. Collectively, these *in silico* approaches reduce the time, cost, and resources required for early-stage drug discovery while providing a rational framework for selecting promising compounds for subsequent molecular dynamics simulations and experimental validation.

1.5 Research Gap

Although numerous computational studies have investigated natural compounds as potential acetylcholinesterase inhibitors, many have primarily focused on molecular docking as a standalone technique for ranking candidate molecules. However, docking scores alone are insufficient for lead prioritization because they do not account for important factors such as physicochemical properties, drug-likeness, pharmacokinetic behavior, and the molecular basis of protein–ligand interactions. Integrating molecular docking with molecular descriptor analysis, statistical evaluation, protein–ligand interaction analysis, and ADMET prediction provides a more comprehensive framework for identifying promising lead compounds before experimental validation. Therefore, there remains a need for reproducible computational workflows that combine multiple *in silico* approaches to improve the reliability of early-stage lead identification. The present study addresses this need by integrating molecular docking with Python-based data analysis, molecular descriptor evaluation, protein–ligand interaction analysis, and pharmacokinetic prediction to prioritize natural compounds targeting human acetylcholinesterase.

1.6 Aim of the Study

The present study aimed to develop an integrated computational pipeline for the virtual screening and prioritization of natural compounds targeting human acetylcholinesterase (PDB ID: 4EY7), a validated therapeutic target for Alzheimer's disease. A total of 41 natural compounds were evaluated using molecular docking with AutoDock Vina implemented in PyRx. The docking results were further investigated through Python-based molecular descriptor analysis, descriptive statistical evaluation, and data visualization. Protein–ligand interactions were analyzed using Discovery Studio Visualizer, while drug-likeness and pharmacokinetic properties were assessed using Lipinski's Rule of Five and the SwissADME web server. By integrating these computational approaches, this study aimed to identify promising lead compounds for future molecular dynamics simulations and experimental validation, while demonstrating a reproducible workflow for early-stage computational drug discovery.

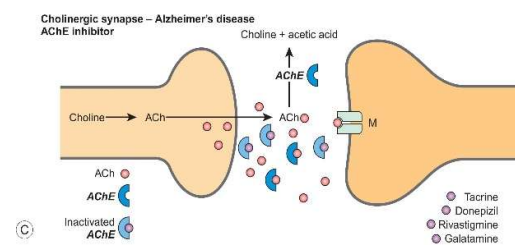
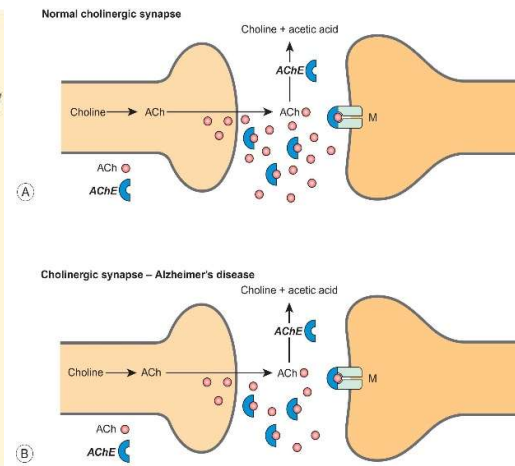
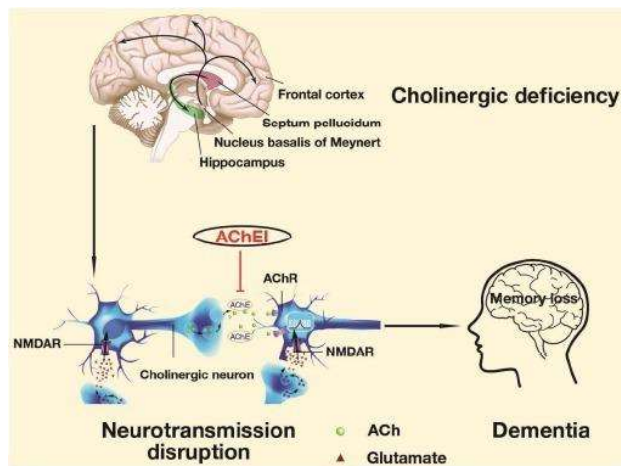


Figure 1. Schematic representation of cholinergic neurotransmission illustrating the role of acetylcholinesterase (AChE) in the hydrolysis of acetylcholine. In Alzheimer's disease, inhibition of AChE increases synaptic acetylcholine levels and improves cholinergic neurotransmission, providing symptomatic relief.

2. Materials and Methods

2.1 Study Design

This study employed an integrated computational workflow to identify and prioritize natural compounds with potential inhibitory activity against human acetylcholinesterase (AChE), a validated therapeutic target for Alzheimer's disease. The overall workflow consisted of sequential stages including target protein selection and preparation, ligand selection and preparation, molecular docking, molecular descriptor analysis, statistical evaluation, protein–ligand interaction analysis, drug-likeness assessment, and pharmacokinetic prediction.

Human acetylcholinesterase (PDB ID: 4EY7) was selected as the target protein because it is co-crystallized with the clinically approved inhibitor donepezil, allowing accurate identification of the active binding site. A total of 41 natural compounds reported to possess neuroprotective, antioxidant, anti-inflammatory, or acetylcholinesterase inhibitory properties were selected through literature review and retrieved from the PubChem database.

Molecular docking was performed using AutoDock Vina implemented in PyRx to predict the binding affinity and binding orientation of each ligand within the active site of acetylcholinesterase. The resulting docking scores were analyzed using Python-based data analytics, including molecular descriptor evaluation, descriptive statistical analysis, and graphical visualization. The highest-ranked compounds were subsequently subjected to protein–ligand interaction analysis using Discovery Studio Visualizer. Drug-likeness was evaluated according to Lipinski's Rule of Five, while pharmacokinetic properties were predicted using the SwissADME web server. The integrated workflow was designed to prioritize promising lead compounds for future computational and experimental validation.

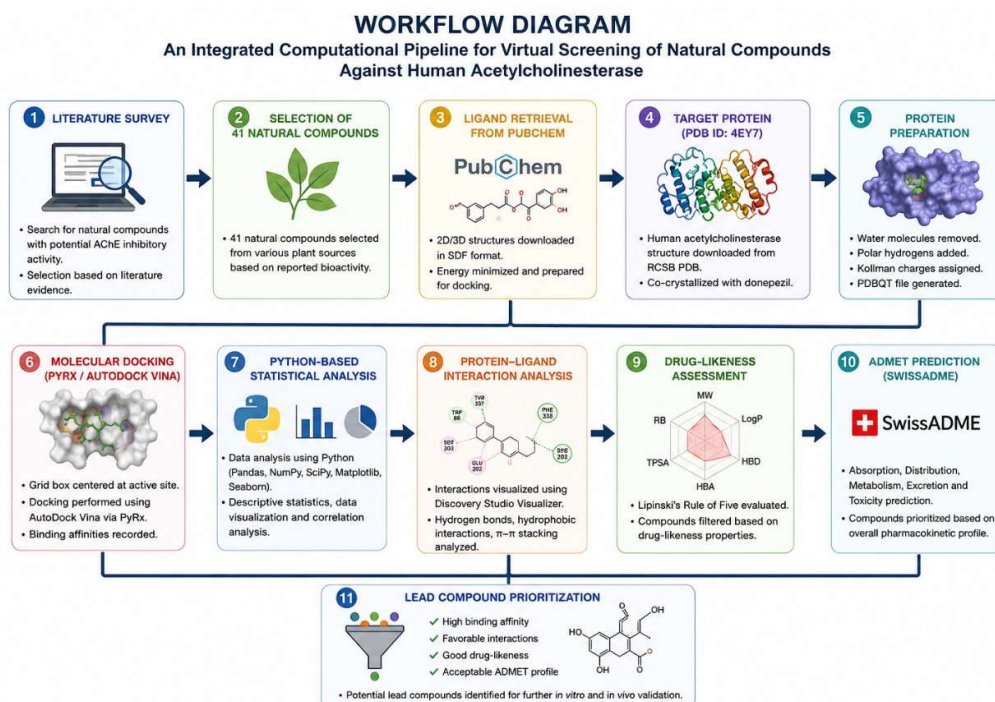


Figure 2 . Overview of the integrated computational workflow employed for the virtual screening and prioritization of natural compounds against human acetylcholinesterase. The workflow includes target and ligand preparation, molecular docking, Python-based data analysis, protein–ligand interaction analysis, drug-likeness assessment, ADMET prediction, and lead compound prioritization.

2.2 Target Protein Selection and Preparation

The three-dimensional crystal structure of human acetylcholinesterase (AChE) was obtained from the Protein Data Bank (PDB ID: 4EY7). This structure was selected because it represents a high-resolution X-ray crystal structure (2.35 Å) co-crystallized with the clinically approved acetylcholinesterase inhibitor donepezil, enabling accurate identification of the enzyme's active binding site for molecular docking.

Prior to docking, the protein structure was prepared using PyRx. The co-crystallized ligand (donepezil) and crystallographic water molecules were removed to eliminate unnecessary molecules that could interfere with ligand binding during docking. The prepared protein structure was then converted into the format required for AutoDock Vina within the PyRx environment.

The active binding site for molecular docking was defined based on the position of the co-crystallized donepezil molecule in the crystal structure. Using a binding site validated by an experimentally determined ligand improves the biological relevance of the docking calculations and ensures that all screened compounds are evaluated within the catalytic gorge of human acetylcholinesterase. The prepared receptor structure was subsequently used for all molecular docking analyses performed in this study.

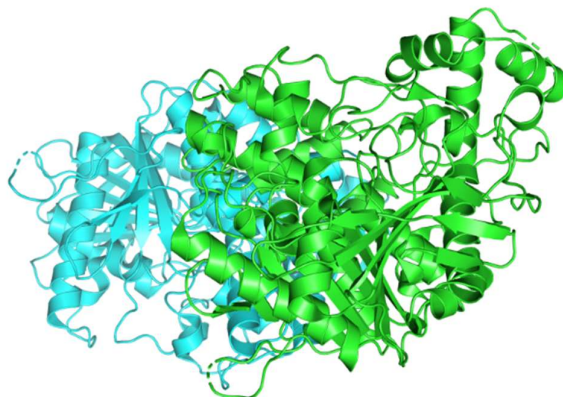


Figure 3. Three-dimensional crystal structure of human acetylcholinesterase (PDB ID: 4EY7) used as the receptor for molecular docking. The co-crystallized inhibitor donepezil (stick representation) indicates the experimentally validated active binding site employed for virtual screening.

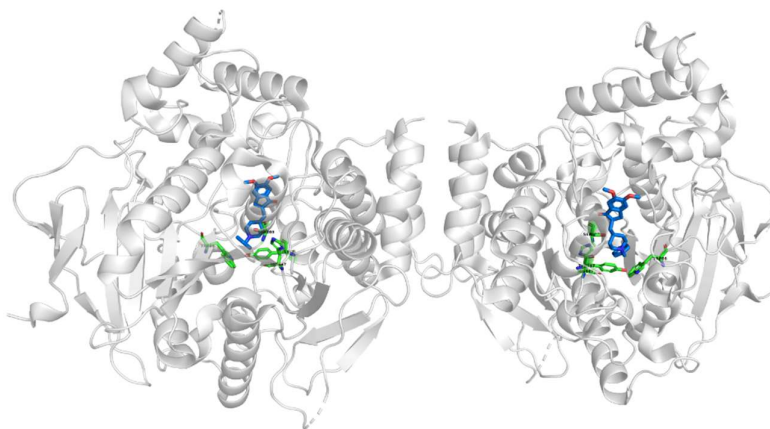


Figure 4. Close-up view of the active binding pocket of human acetylcholinesterase showing the position of the co-crystallized ligand (donepezil) within the catalytic gorge.

Protein Information

Parameter	Value
Protein Name	Human Acetylcholinesterase
PDB ID	4EY7
Organism	Homo sapiens
Experimental Method	X-ray crystallography
Resolution	2.35 Å
Co-crystallized Ligand	Donepezil
Target Class	Hydrolase

2.3 Ligand Selection and Preparation

A total of 41 natural compounds were selected based on an extensive literature survey of peer-reviewed research articles reporting neuroprotective, antioxidant, anti-inflammatory, or acetylcholinesterase inhibitory activities. The selected compounds represented diverse classes of phytochemicals, including flavonoids, polyphenols, alkaloids, terpenoids, phenolic acids, and curcuminoids. Donepezil, a

clinically approved acetylcholinesterase inhibitor, was included as the reference ligand for comparative evaluation of docking performance. The three-dimensional molecular structures and associated chemical information of all selected compounds were obtained from the PubChem database. Canonical SMILES, PubChem Compound Identifiers (CID), and molecular

structures were retrieved for subsequent computational analyses. The downloaded ligand structures were imported into PyRx, where energy minimization was performed using the Open Babel module with the MMFF94 force field to optimize molecular geometry prior to docking. The optimized ligands were then converted into the format required for AutoDock Vina within the PyRx environment.

To ensure consistency throughout the virtual screening process, all ligands were prepared using the same optimization protocol and docked under identical docking conditions. The prepared ligand library was subsequently used for molecular docking against the active site of human acetylcholinesterase.

Selected Ligands Used in This Study

Compound	PubChem CID	Plant Source	Compound Class	Reported Activity
Donepezil	3152	Reference	Reference	Reference
Quercetin	5280343	Onion, Apple	Flavonoid	AChE inhibition, antioxidant
Kaempferol	5280863	Tea, Kale	Flavonoid	Neuroprotective
Myricetin	5281672	Berries	Flavonoid	Anti-amyloid, antioxidant
Luteolin	5280445	Celery	Flavonoid	AChE inhibition
Apigenin	5280443	Chamomile	Flavonoid	Neuroprotective
Naringenin	439246	Citrus fruits	Flavonoid	Anti-inflammatory
Hesperidin	10621	Orange	Flavonoid	Cognitive improvement
Catechin	9064	Green tea	Flavonoid	Antioxidant
Epicatechin	72276	Cocoa	Flavonoid	Neuroprotective
EGCG	65064	Green tea	Flavonoid	Anti-amyloid aggregation
Rutin	5280805	Buckwheat	Flavonoid	AChE inhibition
Fisetin	5281614	Strawberry	Flavonoid	Neuroprotective
Baicalein	5281605	Scutellaria baicalensis	Flavonoid	AChE inhibition
Genistein	5280961	Soybean	Isoflavonoid	Neuroprotective
Chrysin	5281607	Honey, Propolis	Flavonoid	Anti-inflammatory
Curcumin	969516	Turmeric	Polyphenol	Anti-amyloid, antioxidant
Demethoxycurcumin	5469424	Turmeric	Curcuminoid	Neuroprotective
Bisdemethoxycurcumin	5315472	Turmeric	Curcuminoid	Anti-Alzheimer's
Resveratrol	445154	Grapes	Stilbene	Neuroprotective
Pterostilbene	5281727	Blueberries	Stilbene	Antioxidant
Gallic acid	370	Tea	Phenolic acid	AChE inhibition
Caffeic acid	689043	Coffee	Phenolic acid	Antioxidant
Ferulic acid	445858	Rice bran	Phenolic acid	Neuroprotective
Chlorogenic acid	1794427	Coffee	Phenolic acid	Anti-inflammatory
Rosmarinic acid	5281792	Rosemary	Phenolic acid	AChE inhibition
Ellagic acid	5281855	Pomegranate	Phenolic acid	Neuroprotective
Berberine	2353	Berberis spp.	Alkaloid	AChE inhibition
Piperine	638024	Black pepper	Alkaloid	Neuroprotective
Harmine	5280953	Peganum harmala	Alkaloid	AChE inhibition
Huperzine A	854026	Huperzia serrata	Alkaloid	Potent AChE inhibitor
Withanolide A	11294368	Ashwagandha	Steroidal lactone	Neuroprotective
Withaferin A	265237	Ashwagandha	Steroidal lactone	Anti-inflammatory
Ginkgolide B	11973122	Ginkgo biloba	Terpenoid	Neuroprotective
Bilobalide	73581	Ginkgo biloba	Terpenoid	Cognitive enhancement
Ursolic acid	64945	Rosemary	Triterpenoid	Anti-inflammatory
Oleanolic acid	10494	Olive	Triterpenoid	Neuroprotective
Asiatic acid	119034	Centella asiatica	Triterpenoid	Neuroprotective
Thymoquinone	10281	Nigella sativa	Quinone	Antioxidant
Scopoletin	5280460	Morinda citrifolia	Coumarin	AChE inhibition
Umbelliferone	5281426	Apiaceae plants	Coumarin	Antioxidant

The natural compounds included in this study were identified through a comprehensive literature survey of peer-reviewed research articles and review papers available in databases such as PubMed and Google Scholar. The literature search was conducted using combinations of keywords including "*Alzheimer's disease*", "*acetylcholinesterase inhibitor*", "*natural compounds*", "*phytochemicals*", and "*neuroprotective agents*". Compounds were selected based on reported antioxidant, anti-inflammatory, neuroprotective, or acetylcholinesterase inhibitory activities, as well as the availability of their chemical structures in the PubChem database.

Representative chemical structures of selected natural compounds included in the virtual screening study.

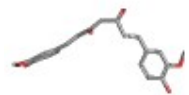
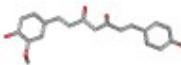
		
Donepezil	EGCG	Curcumin
		
Demethoxycurcumin	Quercetin	Bisdemethoxycurcumin

Figure 5. Representative three-dimensional chemical structures of selected natural compounds and the reference ligand (donepezil) included in the virtual screening study.

2.4 Docking Protocol Validation

Before virtual screening, the molecular docking protocol was validated through a redocking procedure using the co-crystallized ligand donepezil from the human acetylcholinesterase crystal structure (PDB ID: 4EY7). The validation was performed to confirm that the selected docking parameters and binding-site definition were appropriate for reproducing the experimentally observed ligand-binding region.

The co-crystallized ligand was extracted from the receptor structure and prepared using the same ligand preparation protocol applied to the screened natural compounds. Redocking was carried out using AutoDock Vina with a docking grid centered on the original binding site coordinates. An exhaustiveness

value of 16 and a $20 \times 20 \times 20$ Å grid box were employed during the validation procedure to ensure a thorough conformational search.

Following redocking, the predicted binding pose was visually compared with the crystallographic binding orientation within the active site of acetylcholinesterase using BIOVIA Discovery Studio Visualizer. The redocked ligand occupied the same catalytic binding pocket as the experimentally determined ligand and displayed a comparable binding orientation with key active-site residues, indicating that the selected docking protocol was appropriate for subsequent virtual screening.

Based on the successful reproduction of the known binding site and the consistent binding orientation of the reference ligand, the validated docking parameters were subsequently employed for the virtual screening of the selected natural compounds.



Figure 6. Binding orientation of the reference ligand donepezil within the active site of human acetylcholinesterase (PDB ID: 4EY7) following the docking protocol used for validation. The reproduced binding pose supports the suitability of the selected docking parameters for virtual screening.

2.5 Molecular Docking

Molecular docking was performed to evaluate the binding affinity of selected natural compounds toward human acetylcholinesterase (AChE). Docking simulations were carried out using AutoDock Vina through the Python vina package in the Google Colab environment. The prepared receptor structure and ligand library were used as inputs for the virtual screening workflow.

The docking search space was defined based on the centroid of the co-crystallized donepezil molecule present in the crystal structure of human acetylcholinesterase (PDB ID: 4EY7). The docking grid was centered at $X = -14.108$ Å, $Y = -43.833$ Å, and $Z = 27.670$ Å, with grid box dimensions of $24 \times 24 \times 24$ Å, ensuring complete coverage of the active binding pocket. AutoDock Vina affinity maps were generated prior to docking using the same search space for all ligands to maintain consistency throughout the virtual screening process.

Each ligand was docked independently against the prepared receptor using an exhaustiveness value of 8, and nine binding poses were generated for each docking simulation. The binding pose with the lowest predicted binding free energy (kcal/mol) was selected as the representative docking conformation for subsequent analyses. Predicted binding affinity values generated by AutoDock Vina were used to rank

all compounds, with more negative binding energies indicating stronger predicted interactions with the target protein.

The docking results for all screened compounds were automatically exported as comma-separated values (CSV) files for further analysis. Individual docked complexes were also saved in PDBQT format for subsequent protein–ligand interaction analysis using Discovery Studio Visualizer. The resulting docking dataset served as the basis for molecular descriptor analysis, statistical evaluation, drug-likeness assessment, and ADMET prediction performed in the subsequent stages of the computational workflow.

Parameter	Value
Docking Software	AutoDock Vina (Python vina package)
Environment	Google Colab
Receptor	Human AChE (PDB ID: 4EY7)
Grid Center (Å)	(-14.108, -43.833, 27.670)
Grid Box Size (Å)	24 × 24 × 24
Exhaustiveness	8
Number of Binding Poses	9
Output	Binding affinity (kcal/mol)



Figure 7. Docking search space used for virtual screening of natural compounds against human acetylcholinesterase (PDB ID: 4EY7). The grid box was centered on the binding site of the co-crystallized inhibitor donepezil to ensure docking within the catalytic gorge.

2.6 Molecular Descriptor Analysis

Molecular descriptor analysis was performed to investigate the physicochemical properties of the screened natural compounds and to examine their potential relationship with the predicted molecular docking results. Chemical information for each ligand was retrieved programmatically from the PubChem database using the **PubChemPy** library in the Python programming environment. The retrieved data included molecular descriptors commonly employed in drug discovery studies, such as molecular weight (MW), octanol/water partition coefficient (XLogP), topological polar surface area (TPSA), hydrogen bond donor count (HBD), hydrogen bond acceptor count (HBA), rotatable bond count (RB), molecular formula, and PubChem Compound Identifier (CID).

The retrieved descriptor data were imported into **Pandas DataFrames** for preprocessing and quality assessment. Compound names were standardized to ensure consistency across datasets, after which the molecular descriptor dataset was merged with the molecular docking results using ligand names as the common identifier. This generated a unified dataset containing both docking scores and physicochemical properties for all screened compounds.

The integrated dataset served as the basis for subsequent statistical analysis and data visualization performed in Python. Molecular descriptors were used to evaluate the physicochemical characteristics of the screened compounds, investigate trends associated with docking performance, and support the interpretation of drug-likeness and pharmacokinetic analyses presented in later sections of this study.

2.7 Statistical Analysis

Statistical analysis of the molecular docking dataset was performed using **Python** in the **Google Colab** environment to summarize, explore, and visualize the docking results. The molecular docking scores and molecular descriptor dataset were imported into **Pandas DataFrames**, where data preprocessing was carried out to ensure consistency in compound names, remove missing values, and generate a unified dataset for subsequent analyses.

Descriptive statistical analysis was performed to summarize the distribution of docking scores, including the calculation of the mean, median, standard deviation, minimum, maximum, and quartile values. Compounds were ranked according to their predicted binding affinity, with more negative docking scores indicating stronger predicted interactions with human acetylcholinesterase. The ranked dataset facilitated the identification of high-performing compounds for further protein–ligand interaction, drug-likeness, and ADMET analyses.

Exploratory Data Analysis (EDA) was conducted to investigate the overall characteristics of the dataset and identify potential relationships between molecular descriptors and docking performance. Pearson correlation analysis was performed to evaluate associations between docking scores and physicochemical descriptors, including molecular weight, XLogP, topological polar surface area (TPSA), hydrogen bond donor count, hydrogen bond acceptor count, and rotatable bond count. The resulting correlation matrix was visualized as a heatmap to facilitate interpretation of descriptor relationships.

Data visualization was performed using the **Matplotlib** library to generate publication-quality figures, including horizontal bar charts, histograms, box plots, scatter plots, pie charts, and correlation heatmaps. These visualizations were used to examine the distribution of docking scores, compare compound

performance, identify potential outliers, and explore relationships among physicochemical properties and predicted binding affinity.

2.8 Protein–Ligand Interaction Analysis

Protein–ligand interaction analysis was performed to investigate the binding mode of the highest-scoring natural compound (epigallocatechin gallate, EGCG) and the reference inhibitor donepezil within the active site of human acetylcholinesterase (AChE). These two ligands were selected for detailed structural comparison based on their molecular docking results.

The docked protein–ligand complexes generated by AutoDock Vina were imported into BIOVIA Discovery Studio Visualizer (Version 2024) for visualization and interaction analysis. Three-dimensional (3D) binding poses were examined to evaluate ligand orientation within the catalytic gorge of acetylcholinesterase. The amino acid residues located within 4 Å of each ligand were identified to characterize the binding environment and compare interaction patterns between the reference drug and the highest-ranked natural compound.

The interaction analysis focused on identifying residues surrounding the binding pocket and evaluating the binding orientation of the ligands within the active site. The observed interaction profiles were subsequently compared with the molecular docking results to support the selection of promising lead compounds for further drug-likeness and pharmacokinetic analyses.



Figure 8. Three-dimensional binding pose of epigallocatechin gallate (EGCG) within the active site of human acetylcholinesterase (PDB ID: 4EY7). The protein is shown in cartoon representation, while EGCG is displayed as stick representation within the catalytic binding pocket.



Figure 9. Three-dimensional binding pose of the reference inhibitor donepezil within the active site of human acetylcholinesterase (PDB ID: 4EY7), illustrating its binding orientation within the catalytic gorge.

Ligand	BindingEnergy (kcal/mol)	Key Residues (within 4 Å)
EGCG	-11.972	Tyr72, Asp74, Trp86, Tyr124, Ser203, Trp286, Tyr337, Phe338, Tyr341, His447
Donepezil	-11.771	Tyr72, Trp86, Tyr124, Glu202, Trp286, Tyr337, Phe338, Tyr341, His447, Gly448

2.9 Drug-Likeness Assessment

The drug-likeness of the screened natural compounds was evaluated using **Lipinski's Rule of Five**, a widely accepted guideline for assessing the oral drug-likeness of small molecules. The analysis was performed using **Python** in the **Google Colab** environment based on the molecular descriptors retrieved from the PubChem database.

The evaluated parameters included **molecular weight ($MW \leq 500$ Da)**, **octanol/water partition coefficient ($XLogP \leq 5$)**, **hydrogen bond donors ($HBD \leq 5$)**, and **hydrogen bond acceptors ($HBA \leq 10$)**. Each compound was assessed against these criteria to determine the number of rule violations. Compounds satisfying all four criteria were considered to exhibit favorable drug-like characteristics according to Lipinski's guidelines.

The molecular descriptor dataset generated during the descriptor analysis was used as the input for the automated evaluation. Python scripts were employed to compare each molecular descriptor against the corresponding Lipinski threshold and to classify compounds according to their compliance. The resulting drug-likeness assessment was subsequently integrated with the molecular docking results and ADMET predictions to support the prioritization of potential lead compounds for further investigation.

Parameter	Acceptable Range
Molecular Weight	≤ 500 Da
XLogP	≤ 5
Hydrogen Bond Donors	≤ 5
Hydrogen Bond Acceptors	≤ 10

2.10 Pharmacokinetic Property Prediction Using SwissADME

Pharmacokinetic and drug-likeness properties of the highest-ranked compounds were evaluated using the **SwissADME** web server to assess their suitability as potential drug candidates. Following molecular docking, the top ten compounds with the most favorable predicted binding affinities were selected for ADMET analysis.

Canonical SMILES for the selected compounds were retrieved programmatically from the **PubChem** database using the **PubChemPy** library in Python. The retrieved SMILES strings were compiled into a standardized dataset and exported as a comma-separated values (CSV) file to facilitate batch submission to the SwissADME web server.

SwissADME was used to predict key physicochemical, pharmacokinetic, and medicinal chemistry properties, including gastrointestinal (GI) absorption, blood–brain barrier (BBB) permeability, Lipinski's Rule of Five compliance, bioavailability score, PAINS (Pan-Assay Interference Compounds) alerts, Brenk structural alerts, and synthetic accessibility. These parameters were evaluated to determine the potential oral bioavailability and overall drug-likeness of the selected compounds.

The predicted ADMET properties were subsequently integrated with the molecular docking results and Lipinski's Rule of Five assessment to prioritize compounds exhibiting favorable binding affinity together with acceptable pharmacokinetic characteristics. This integrated evaluation enabled a more comprehensive selection of potential lead compounds for future computational and experimental investigations.

Parameter	Description
GI Absorption	Predicted intestinal absorption
BBB Permeability	Ability to cross the blood–brain barrier
Lipinski Compliance	Drug-likeness assessment
Bioavailability Score	Predicted oral bioavailability
PAINS Alerts	Detection of assay-interfering substructures
Brenk Alerts	Identification of potentially problematic structural fragments
Synthetic Accessibility	Estimated ease of chemical synthesis

2.11 Software and Computational Environment

All computational analyses were performed using **Python** within the **Google Colab** cloud-based computing environment. Molecular docking was conducted using the **AutoDock Vina** docking engine through the Python **vina** package, while ligand preparation was carried out using the **Meeko** toolkit and **RDKit**. Molecular descriptors and chemical information were retrieved programmatically from the **PubChem** database using the **PubChemPy** library.

Data preprocessing, statistical analysis, and exploratory data analysis were performed using the **Pandas** and **NumPy** libraries. Graphical representations of the docking results and molecular descriptor analyses were generated using **Matplotlib**. Protein–ligand complexes were visualized and analyzed using **BIOVIA Discovery Studio Visualizer 2024**, while pharmacokinetic and drug-likeness predictions were obtained using the **SwissADME** web server.

The complete computational workflow was implemented using reproducible Python scripts, enabling automated data retrieval, preprocessing, statistical analysis, and result generation while minimizing manual intervention and improving the reproducibility of the study.

Software / Tool	Purpose
Google Colab	Cloud-based computational environment
Python	Data processing and workflow automation
AutoDock Vina	Molecular docking
Vina (Python package)	Docking implementation
Meeko	Ligand preparation and PDBQT generation
RDKit	Molecular processing and cheminformatics
PubChemPy	Retrieval of molecular descriptors and SMILES
Pandas	Data preprocessing and analysis
NumPy	Numerical computations
Matplotlib	Data visualization
BIOVIA Discovery Studio Visualizer 2024	Protein–ligand interaction analysis
SwissADME	Pharmacokinetic and drug-likeness prediction

3.Results and Discussion

3.1 Molecular Docking Results

Molecular docking was performed to evaluate the binding affinity of 41 natural compounds against human acetylcholinesterase (AChE). The predicted binding affinities obtained using AutoDock Vina ranged from -11.972 to -5.506 kcal/mol, indicating considerable variation in the binding potential of the screened compounds. More negative binding energy values represent stronger predicted interactions between the ligand and the target protein.

Among all screened compounds, epigallocatechin gallate (EGCG) exhibited the strongest predicted binding affinity (-11.972 kcal/mol), marginally exceeding that of the reference inhibitor donepezil (-11.771 kcal/mol). Other high-ranking compounds included demethoxycurcumin (-10.948 kcal/mol), followed by additional natural compounds with favorable docking scores, suggesting their potential to interact effectively with the active site of human acetylcholinesterase.

The docking results demonstrated that several polyphenolic compounds, particularly flavonoids and curcuminoids, exhibited strong predicted binding affinities. These compounds possess multiple hydroxyl groups and aromatic ring systems that may facilitate hydrogen bonding and π -mediated interactions within the catalytic gorge of acetylcholinesterase. Nevertheless, docking scores alone do not establish biological activity or therapeutic efficacy and should be interpreted alongside molecular interaction analysis, drug-likeness assessment, and pharmacokinetic predictions.

Overall, the molecular docking analysis successfully identified several natural compounds with predicted binding affinities comparable to or exceeding that of the reference drug donepezil, providing a rational basis for their further computational evaluation.

Docking Result

Ligand	Binding Energy (kcal/mol)	Ligand	Binding Energy (kcal/mol)
EGCG	-11.972	Myricetin	-10.023
Donepezil	-11.771	Epicatechin	-9.955
Demethoxycurcumin	-10.948	Catechin	-9.907
Curcumin	-10.932	Chlorogenic acid	-9.905
Bisdemethoxycurcumin	-10.749	Huperzine A	-9.84
Hesperidin	-10.739	Rutin	-9.758
Rosmarinic acid	-10.675	Genistein	-9.634
Baicalein	-10.57	Ursolic acid	-9.591
Ellagic acid	-10.447	Pterostilbene	-9.331
Chrysin	-10.411	Bilobalide	-9.301
Withanolide A	-10.37	Kaempferol	-9.3
Luteolin	-10.343	Resveratrol	-9.246
Piperine	-10.33	Ginkgolide B	-8.943
Apigenin	-10.311	Harmine	-8.816
Naringenin	-10.276	Oleanolic acid	-8.288
Berberine	-10.273	Asiatic acid	-7.964
Quercetin	-10.183	Umbelliferone	-7.729
Withaferin A	-10.107	Scopoletin	-7.649
Fisetin	-10.092	Ferulic acid	-7.56
		Caffeic acid	-7.544

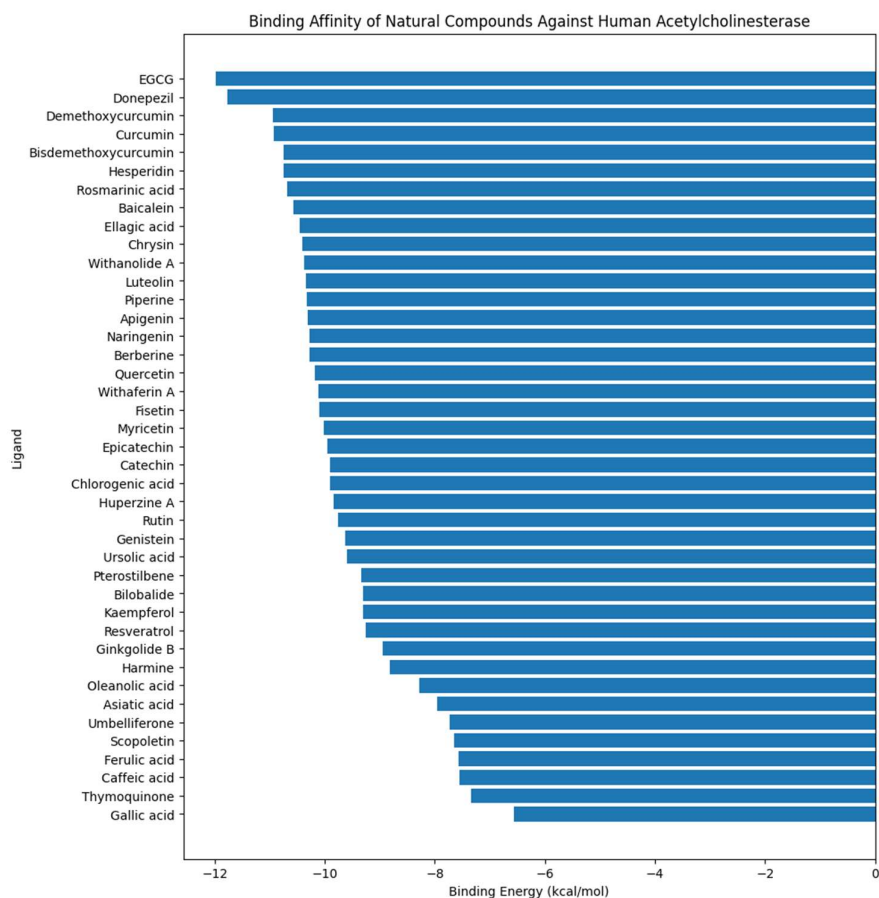


Figure 10 Predicted binding affinities of the screened natural compounds and the reference inhibitor donepezil against human acetylcholinesterase obtained using AutoDock Vina.

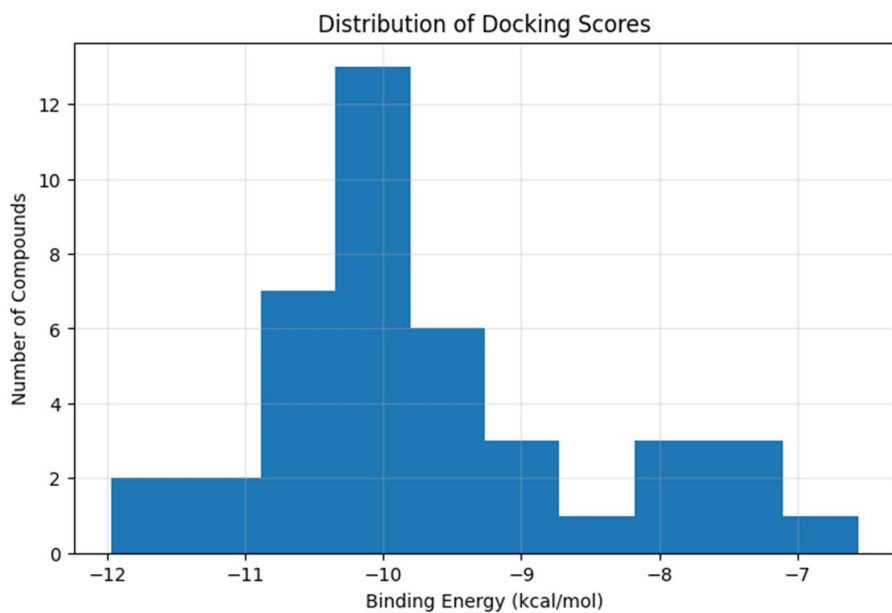


Figure 11. Distribution of predicted molecular docking binding affinities of the 41 screened natural compounds against human acetylcholinesterase (PDB ID: 4EY7). The histogram illustrates the frequency distribution of binding energies obtained using AutoDock Vina, with more negative values indicating stronger predicted protein–ligand interactions.

3.2 Molecular Descriptor Analysis

Molecular descriptor analysis was performed to characterize the physicochemical properties of the screened natural compounds and to investigate their potential relationship with molecular docking performance. Descriptors including molecular weight (MW), octanol/water partition coefficient (XLogP), topological polar surface area (TPSA), hydrogen bond donor count (HBD), hydrogen bond acceptor count (HBA), and rotatable bond count (RB) were retrieved from the PubChem database and integrated with the docking dataset.

Considerable variation was observed among the screened compounds with respect to their physicochemical properties, reflecting the structural diversity of the selected natural products. Several high-ranking compounds, including **epigallocatechin gallate (EGCG)** and the curcuminoids, possessed relatively high molecular weights and multiple hydrogen bond donor and acceptor groups. These structural characteristics increase the potential for forming stabilizing hydrogen-bonding interactions with amino acid residues within the active site of human acetylcholinesterase, which may contribute to their favorable predicted binding affinities.

The descriptor analysis also demonstrated substantial differences in lipophilicity (XLogP), molecular flexibility, and molecular polarity among the screened compounds. Such variations are important because these properties influence not only protein–ligand interactions but also pharmacokinetic behavior, including membrane permeability and oral bioavailability. Consequently, molecular descriptor analysis provided complementary information to the docking results and established a physicochemical basis for subsequent drug-likeness assessment, correlation analysis, and pharmacokinetic prediction.

Overall, the integration of molecular descriptors with docking scores enabled a more comprehensive evaluation of the screened compounds by considering both predicted target affinity and physicochemical characteristics relevant to early-stage drug discovery.

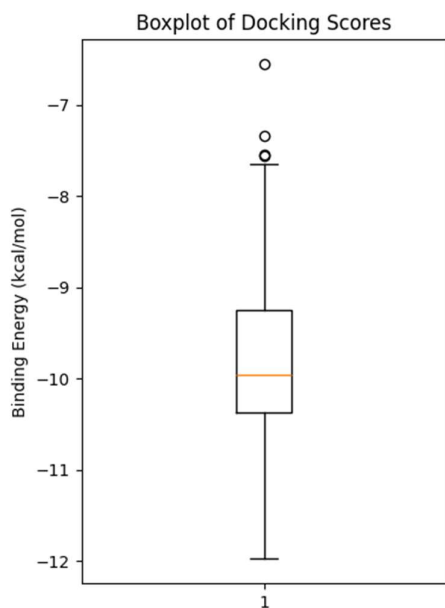


Figure 12 Box plot illustrating the distribution of predicted molecular docking binding affinities of the 41 screened natural compounds against human acetylcholinesterase (PDB ID: 4EY7). The central line represents the median binding energy, the box indicates the interquartile range (IQR), the whiskers represent the overall data spread excluding outliers, and individual points denote potential outliers. More negative binding energy values correspond to stronger predicted protein–ligand interactions.

CORRELATION MATRIX

Ligand	PubChem CID	Molecular Weight	XLogP	TPSA	H Bond Donor	H Bond Acceptor	Rotatable Bonds
EGCG	65064	458.4	1.2	197	8	11	4
Donepezil	3152	379.5	4.3	38.8	0	4	6
Demethoxycurcumin	5469424	338.4	3.3	83.8	2	5	7
Curcumin	969516	368.4	3.2	93.1	2	6	8
Bisdemethoxycurcumin	5315472	308.3	3.3	74.6	2	4	6
Hesperidin	10621	610.6	-1.1	234	8	15	7
Rosmarinic acid	5281792	360.3	2.4	145	5	8	7
Baicalein	5281605	270.24	1.7	87	3	5	1
Ellagic acid	5281855	302.19	1.1	134	4	8	0
Chrysin	5281607	254.24	2.1	66.8	2	4	1
Withanolide A	11294368	470.6	3.3	96.4	2	6	2
Luteolin	5280445	286.24	1.4	107	4	6	1
Piperine	638024	285.34	3.5	38.8	0	3	3
Apigenin	5280443	270.24	1.7	87	3	5	1
Naringenin	439246	272.25	2.4	87	3	5	1
Berberine	2353	336.4	3.6	40.8	0	4	2
Quercetin	5280343	302.23	1.5	127	5	7	1
Withaferin A	265237	470.6	3.8	96.4	2	6	3
Fisetin	5281614	286.24	2	107	4	6	1
Myricetin	5281672	318.23	1.2	148	6	8	1
Epicatechin	72276	290.27	0.4	110	5	6	1
Catechin	9064	290.27	0.4	110	5	6	1
Chlorogenic acid	1794427	354.31	-0.4	165	6	9	5
Huperzine A	854026	242.32	0	55.1	2	2	0
Rutin	5280805	610.5	-1.3	266	10	16	6
Genistein	5280961	270.24	2.7	87	3	5	1
Ursolic acid	64945	456.7	7.3	57.5	2	3	1
Pterostilbene	5281727	256.3	3.8	38.7	1	3	4
Bilobalide	73581	326.3	-0.3	119	2	8	1
Kaempferol	5280863	286.24	1.9	107	4	6	1
Resveratrol	445154	228.24	3.1	60.7	3	3	2
Ginkgolide B	11973122	424.4	-0.4	149	3	10	1
Harmin	5280953	212.25	3.6	37.9	1	2	1
Oleanolic acid	10494	456.7	7.5	57.5	2	3	1
Asiatic acid	119034	488.7	5.7	98	4	5	2
Umbelliferone	5281426	162.14	1.6	46.5	1	3	0
Scopoletin	5280460	192.17	1.5	55.8	1	4	1
Ferulic acid	445858	194.18	1.5	66.8	2	4	3
Caffeic acid	689043	180.16	1.2	77.8	3	4	2
Thymoquinone	10281	164.2	2	34.1	0	2	1
Gallic acid	370	170.12	0.7	98	4	5	1

3.3 Exploratory Data Analysis and Correlation Analysis

Exploratory Data Analysis (EDA) was performed to investigate the distribution of molecular descriptors and to examine potential relationships between the physicochemical properties of the screened compounds and their predicted binding affinities. The integrated dataset consisting of molecular docking scores and molecular descriptors was analyzed using Python, and graphical visualization techniques were employed to identify trends, correlations, and potential outliers.

Pearson correlation analysis demonstrated that the molecular docking score exhibited a **moderate negative correlation** with molecular weight ($r = -0.435$) and the number of rotatable bonds ($r = -0.444$), indicating that compounds with higher molecular weight or greater molecular flexibility generally tended to display more favorable (more negative) predicted binding affinities within this dataset. In contrast, the docking score showed **almost no correlation** with lipophilicity (XLogP; $r = 0.010$), suggesting that hydrophobicity alone was not a major determinant of docking performance among the screened natural compounds.

The correlation matrix also revealed expected relationships among several molecular descriptors. For example, molecular weight exhibited positive correlations with topological polar surface area (TPSA), hydrogen bond donor count (HBD), hydrogen bond acceptor count (HBA), and the number of rotatable bonds, reflecting the structural characteristics of larger and more chemically complex molecules. These findings indicate that multiple physicochemical properties collectively influence ligand binding rather than any single descriptor acting independently.

Overall, the exploratory analysis demonstrated that molecular docking performance is governed by a combination of molecular size, flexibility, polarity, and hydrogen-bonding capacity. These observations further support the importance of integrating molecular descriptor analysis with docking results when prioritizing lead compounds for subsequent pharmacokinetic evaluation.

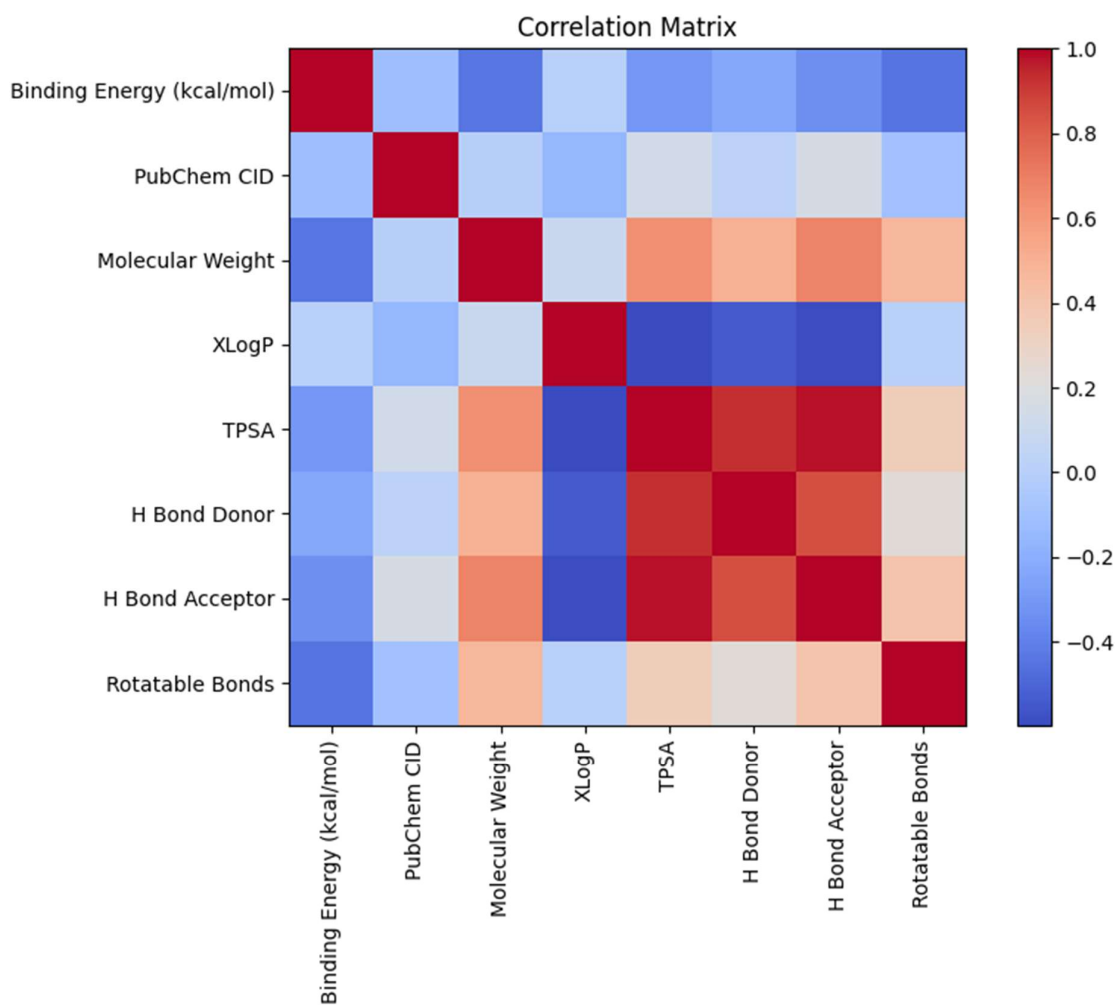


Figure 13. Pearson correlation matrix illustrating the relationships between predicted molecular docking binding affinity and selected molecular descriptors of the screened natural compounds. The heatmap displays pairwise correlation coefficients among binding energy, molecular weight, XLogP, topological polar surface area (TPSA), hydrogen bond donor count (HBD), hydrogen bond acceptor count (HBA), and rotatable bond count. Warmer colors indicate positive correlations, whereas cooler colors indicate negative correlations.

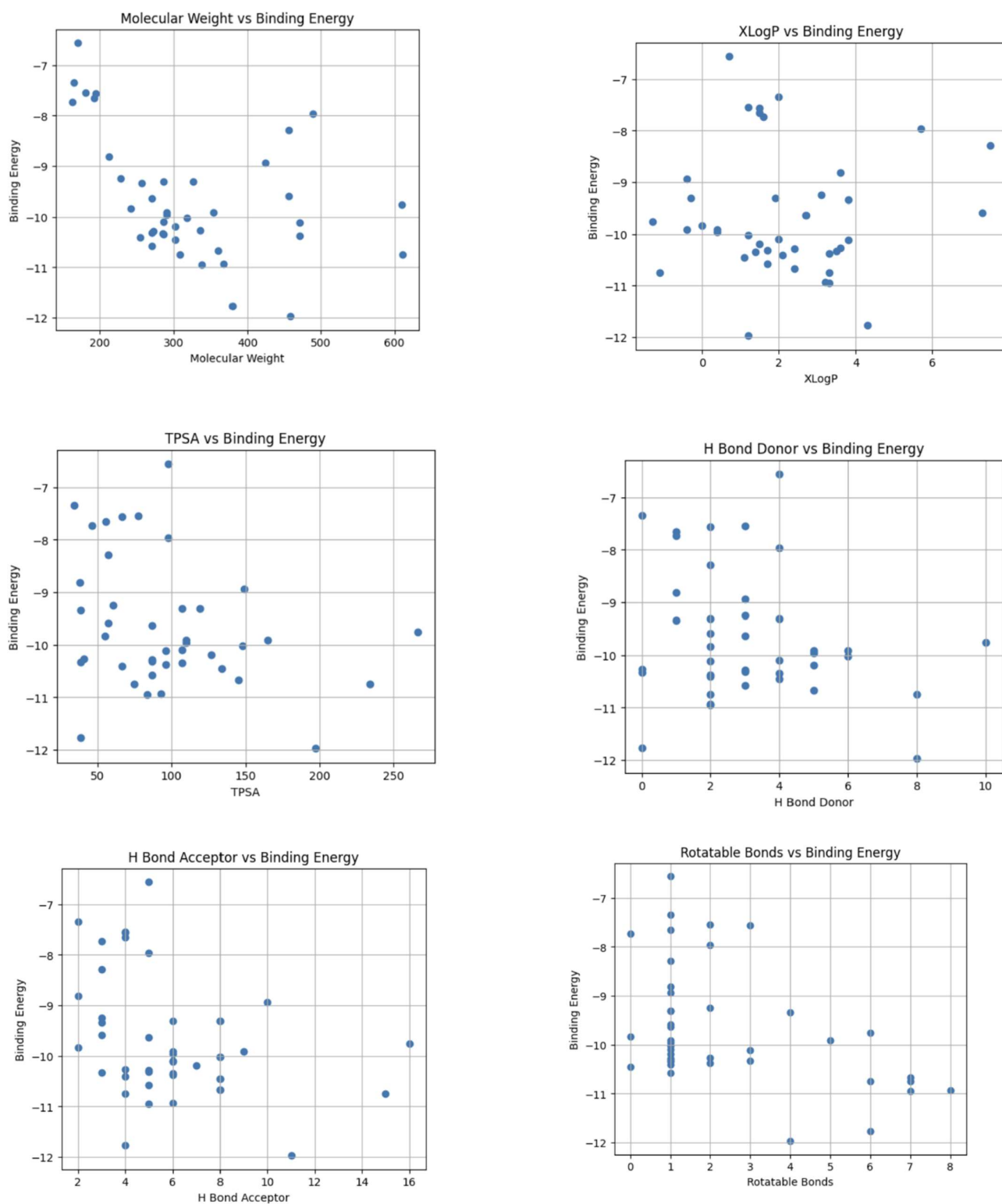


Figure 14 Scatter plots illustrating the relationships between predicted molecular docking binding affinity and selected molecular descriptors of the screened natural compounds. The analyses compare binding energy with molecular weight, XLogP, topological polar surface area (TPSA), hydrogen bond donor count (HBD), hydrogen bond acceptor count (HBA), and rotatable bond count. These plots were generated to explore potential associations between physicochemical properties and predicted protein–ligand binding affinity.

3.4 Protein–Ligand Interaction Analysis

To investigate the molecular basis of ligand binding, detailed protein–ligand interaction analysis was performed for the highest-ranked natural compound, **epigallocatechin gallate (EGCG)**, and the reference inhibitor **donepezil**. The docked complexes were visualized using **BIOVIA Discovery Studio Visualizer 2024**, allowing examination of ligand orientation within the active site of human acetylcholinesterase (AChE) and identification of amino acid residues surrounding the binding pocket.

EGCG exhibited the strongest predicted binding affinity among all screened natural compounds (**−11.972 kcal/mol**). Structural visualization revealed that EGCG occupied the catalytic gorge of acetylcholinesterase and interacted with several key residues, including **Tyr72, Asp74, Trp86, Tyr124, Ser203, Trp286, Tyr337, Phe338, Tyr341, and His447**. The presence of multiple hydroxyl groups within the EGCG structure provides opportunities for hydrogen-bond formation, while its aromatic rings may facilitate π -mediated interactions with aromatic residues lining the active site. These complementary interactions are consistent with its highly favorable predicted binding affinity.

The reference inhibitor **donepezil** exhibited a predicted binding affinity of **−11.771 kcal/mol** and occupied a similar region within the catalytic gorge. The docked complex showed interactions with several important residues, including **Tyr72, Trp86, Tyr124, Glu202, Trp286, Tyr337, Phe338, Tyr341, His447, and Gly448**. The overlap in binding-site residues between donepezil and EGCG suggests that both ligands target the same functional region of acetylcholinesterase despite their distinct chemical structures.

Comparison of the two binding modes demonstrated that EGCG and donepezil shared interactions with several catalytically important residues, particularly **Trp86, Tyr337, Phe338, Tyr341, Trp286, and His447**, indicating that EGCG adopts a binding orientation comparable to that of the clinically approved inhibitor. These findings support the molecular docking results and provide structural evidence for the favorable predicted interaction of EGCG with human acetylcholinesterase. However, these observations are based on computational predictions and require experimental validation to confirm the proposed binding mechanism.

Parameter	EGCG	Donepezil
BindingEnergy (kcal/mol)	−11.972	−11.771
Common Residues	Trp86,Trp286,Tyr337,Phe338, Tyr341, His447	Trp86,Trp286,Tyr337,Phe338,Tyr341, His447
Unique Residues	Asp74, Ser203	Glu202, Gly448
Binding Pocket	Catalytic gorge	Catalytic gorge

3.5 Drug-Likeness Assessment

The drug-likeness of the screened compounds was evaluated using **Lipinski's Rule of Five**, which provides a preliminary assessment of the suitability of small molecules for oral drug development. The analysis considered molecular weight, lipophilicity (XLogP), hydrogen bond donor count (HBD), and hydrogen bond acceptor count (HBA) to identify compounds with physicochemical properties consistent with favorable oral bioavailability.

The evaluation revealed that the majority of the screened compounds complied with Lipinski's criteria, indicating generally favorable drug-like characteristics. However, several highly ranked natural compounds exhibited one or more rule violations, primarily due to increased molecular weight and a high number of hydrogen bond donor and acceptor groups. In particular, **epigallocatechin gallate (EGCG)**, despite exhibiting the strongest predicted binding affinity, exceeded the recommended limits for hydrogen bonding capacity, reflecting its highly polar chemical structure.

In contrast, several curcuminoids and other natural compounds demonstrated favorable docking scores while exhibiting fewer or no Lipinski rule violations, suggesting a more balanced combination of target affinity and physicochemical properties. These findings highlight the importance of evaluating drug-likeness alongside molecular docking, as compounds with excellent predicted binding affinity may still present challenges related to oral bioavailability and pharmacokinetic performance.

Overall, the Lipinski analysis provided an additional level of compound prioritization by identifying candidates that combined favorable predicted target binding with acceptable physicochemical characteristics suitable for further computational evaluation.

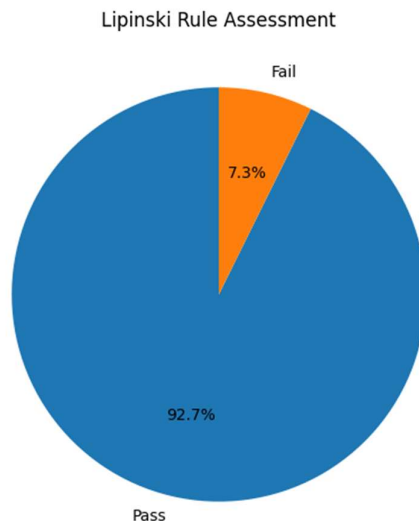


Figure 15 Distribution of the screened natural compounds according to their compliance with Lipinski's Rule of Five. The majority of compounds (92.7%) satisfied the drug-likeness criteria, whereas 7.3% exhibited one or more rule violations, indicating generally favorable physicochemical properties for early-stage drug discovery.

3.6 Pharmacokinetic and Drug-Likeness Prediction

To further evaluate the suitability of the highest-ranked compounds as potential drug candidates, pharmacokinetic and drug-likeness properties were predicted using the **SwissADME** web server. The top ten compounds identified through molecular docking were selected for analysis, and their canonical SMILES were retrieved from the PubChem database for submission to SwissADME.

The prediction results revealed considerable variation in the pharmacokinetic characteristics of the selected compounds. While several compounds exhibited strong predicted binding affinities toward human acetylcholinesterase, not all demonstrated equally favorable pharmacokinetic profiles. Differences were observed in gastrointestinal (GI) absorption, blood–brain barrier (BBB) permeability, bioavailability score, Lipinski compliance, PAINS alerts, Brenk structural alerts, and synthetic accessibility.

Among the evaluated compounds, the reference inhibitor **donepezil** demonstrated a favorable pharmacokinetic profile consistent with its established clinical use. In contrast, **epigallocatechin gallate (EGCG)**, despite exhibiting the strongest predicted binding affinity, showed physicochemical characteristics that may limit its oral bioavailability because of its high polarity and multiple hydrogen-bonding groups. Several curcuminoid derivatives demonstrated a more balanced combination of predicted binding affinity and pharmacokinetic properties, suggesting their potential as promising lead compounds for further investigation.

These findings demonstrate that molecular docking alone is insufficient for lead selection. Integrating pharmacokinetic prediction with docking results enables a more comprehensive evaluation of candidate molecules by considering both target binding and properties relevant to drug absorption and distribution. Consequently, compounds exhibiting an appropriate balance between predicted binding affinity and favorable pharmacokinetic characteristics represent more suitable candidates for future computational optimization and experimental validation.

ADMET RESULT

Rank	Compound	GI Absorption	BBB Permeant	Bioavailability Score	Lipinski Violations	PAINS Alerts	Brenk Alerts
1	EGCG	Low	No	0.17	2	1	1
2	Donepezil	High	Yes	0.55	0	0	0
3	Demethoxycurcumin	High	No	0.55	0	0	2
4	Curcumin	High	No	0.55	0	0	2
5	Bisdemethoxycurcumin	High	Yes	0.55	0	0	2
6	Quercetin	Low	No	0.17	3	0	0
7	Myricetin	Low	No	0.56	0	1	2
8	Kaempferol	High	No	0.55	0	1	1
9	Luteolin	High	No	0.55	0	1	3
10	Apigenin	High	Yes	0.55	0	0	0

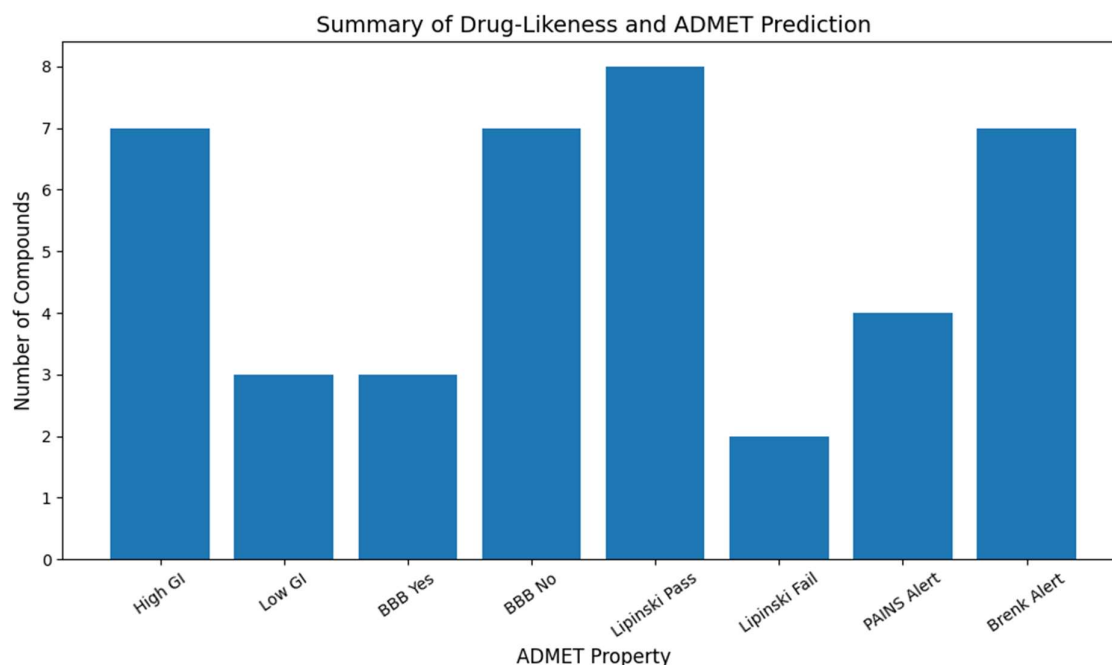


Figure 16 Distribution of predicted ADMET-related properties for the highest-ranked natural compounds evaluated using SwissADME. The analysis summarizes gastrointestinal absorption, blood–brain barrier permeability, Lipinski's Rule of Five compliance, and medicinal chemistry alerts (PAINS and Brenk), facilitating the assessment of compound drug-likeness and pharmacokinetic suitability.

3.7 Lead Compound Prioritization

The identification of promising lead compounds requires the integration of multiple computational analyses rather than relying solely on molecular docking scores. In the present study, lead prioritization was performed by collectively evaluating molecular docking results, molecular descriptors, protein–ligand interaction profiles, Lipinski's Rule of Five compliance, and pharmacokinetic predictions obtained from the SwissADME web server.

Among all screened compounds, **epigallocatechin gallate (EGCG)** exhibited the strongest predicted binding affinity toward human acetylcholinesterase (-11.972 kcal/mol) and demonstrated interactions with several catalytically important residues within the active site. These findings suggest that EGCG possesses strong potential for target recognition and enzyme binding. However, the Lipinski and pharmacokinetic analyses indicated that its high molecular polarity and multiple hydrogen-bonding groups may reduce its predicted oral bioavailability, highlighting potential challenges for its direct development as an orally administered drug candidate.

In contrast, **demethoxycurcumin**, **curcumin**, and **bisdemethoxycurcumin** exhibited slightly lower predicted binding affinities while demonstrating more balanced physicochemical and pharmacokinetic properties. This combination of favorable target binding, acceptable drug-likeness, and improved predicted pharmacokinetic characteristics suggests that these compounds may represent attractive lead candidates for further optimization and experimental investigation.

Overall, the integrated computational workflow demonstrated that effective lead identification should consider both target-binding affinity and drug-like properties. Based on the combined evaluation,

EGCG emerged as the strongest predicted binder to human acetylcholinesterase, whereas the curcuminoid derivatives demonstrated a more balanced profile that may be advantageous for future drug development. These findings provide a rational basis for selecting candidate molecules for subsequent molecular dynamics simulations, binding free-energy calculations, and experimental validation.

Integrated Lead Prioritization of the Top Candidate Compounds

Compound	Docking	Protein Interaction	Lipinski	SwissADME	Overall Assessment
EGCG	Excellent	Excellent	Moderate	Moderate	Strongbinder; pharmacokinetic limitations
Demethoxycurcumin	Very Good	Good	Good	Good	Promising lead candidate
Curcumin	Very Good	Good	Good	Good	Promising lead candidate
Bisdemethoxycurcumin	Very Good	Good	Good	Good	Promising lead candidate
Donepezil	Reference	Reference	Reference	Reference	Approved comparator

4.Limitations

The present study provides an integrated computational evaluation of natural compounds targeting human acetylcholinesterase; however, several limitations should be acknowledged. First, the molecular docking results represent **predicted** protein–ligand interactions based on a static crystal structure and do not fully account for protein flexibility or the dynamic behavior of ligand binding under physiological conditions.

Second, although molecular docking, molecular descriptor analysis, drug-likeness assessment, and pharmacokinetic prediction provide valuable insights for early-stage lead identification, these computational approaches cannot substitute for experimental validation. The predicted binding affinities and pharmacokinetic properties require confirmation through **in vitro** enzymatic assays, **cell-based studies**, and **in vivo** investigations before any conclusions regarding biological efficacy or therapeutic potential can be established.

Third, while pharmacokinetic properties were evaluated using the SwissADME web server, these predictions are based on computational models and should be interpreted as preliminary estimates rather than experimentally verified characteristics. Similarly, compliance with Lipinski's Rule of Five provides only an initial assessment of oral drug-likeness and does not guarantee successful drug development.

Finally, molecular dynamics (MD) simulations and binding free-energy calculations (e.g., MM/PBSA or MM/GBSA) were not performed in the present study. These approaches could provide additional insights into the stability and dynamic behavior of the protein–ligand complexes and further strengthen the computational evaluation of the identified lead compounds.

5.Future Perspectives

The integrated computational workflow presented in this study provides a foundation for the continued investigation of natural compounds as potential acetylcholinesterase inhibitors for Alzheimer's disease. Future research should focus on validating the computational findings through complementary computational techniques and experimental studies to improve confidence in the identified lead compounds.

As an immediate extension of this work, **molecular dynamics (MD) simulations** should be performed to evaluate the stability of the protein–ligand complexes under physiological conditions. Dynamic analyses, together with binding free-energy calculations such as **MM/PBSA** or **MM/GBSA**, would provide a more comprehensive understanding of ligand binding stability and complement the molecular docking results.

The highest-priority compounds identified in this study, particularly **epigallocatechin gallate (EGCG)**, **demethoxycurcumin**, **curcumin**, and **bisdemethoxycurcumin**, should be further investigated through **in vitro acetylcholinesterase inhibition assays** to verify their predicted biological activity. Additional studies examining neuroprotective effects, cytotoxicity, and blood–brain barrier permeability would further support their potential as therapeutic candidates for Alzheimer's disease.

Future work may also expand the computational pipeline by screening larger libraries of natural compounds, incorporating additional pharmacokinetic and toxicity prediction tools, and applying machine learning–based virtual screening approaches to improve lead prioritization. Such integrated computational and experimental strategies may accelerate the discovery and optimization of novel acetylcholinesterase inhibitors for the treatment of Alzheimer's disease.

6. Conclusion

The present study developed and implemented an integrated computational pipeline to identify and prioritize natural compounds with potential inhibitory activity against human acetylcholinesterase (AChE), a validated therapeutic target for Alzheimer's disease. By combining molecular docking, molecular descriptor analysis, exploratory data analysis, protein–ligand interaction analysis, drug-likeness assessment, and pharmacokinetic prediction, the study provided a comprehensive framework for the early-stage evaluation of natural compounds.

Among the 41 screened compounds, **epigallocatechin gallate (EGCG)** exhibited the strongest predicted binding affinity toward human acetylcholinesterase, surpassing the reference inhibitor **donepezil** in the molecular docking simulations. Protein–ligand interaction analysis further demonstrated that EGCG interacted with several key residues within the catalytic gorge of acetylcholinesterase, supporting its favorable predicted binding mode. However, the integrated evaluation also showed that high docking affinity alone is insufficient for lead selection, as pharmacokinetic properties and drug-likeness must also be considered. Curcuminoid derivatives, including **demethoxycurcumin**, **curcumin**, and **bisdemethoxycurcumin**, demonstrated a more balanced combination of predicted binding affinity, physicochemical properties, and pharmacokinetic characteristics, highlighting their potential as promising lead candidates for further investigation.

Overall, this study demonstrates the value of integrating multiple computational approaches to improve the identification and prioritization of potential lead compounds during the early stages of drug discovery. Although the findings are based on computational predictions and require further validation through molecular dynamics simulations and experimental studies, the developed workflow provides a reproducible and systematic approach for computational screening of natural products. The results contribute to ongoing efforts in the discovery of novel acetylcholinesterase inhibitors and establish a solid foundation for future computational and experimental investigations aimed at developing improved therapeutic strategies for Alzheimer's disease.

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Git Hub : <https://github.com/sayan7585das-cmyk/sayan7585das-cmyk>

Linkdin : www.linkedin.com/in/sayan-das-19660340a